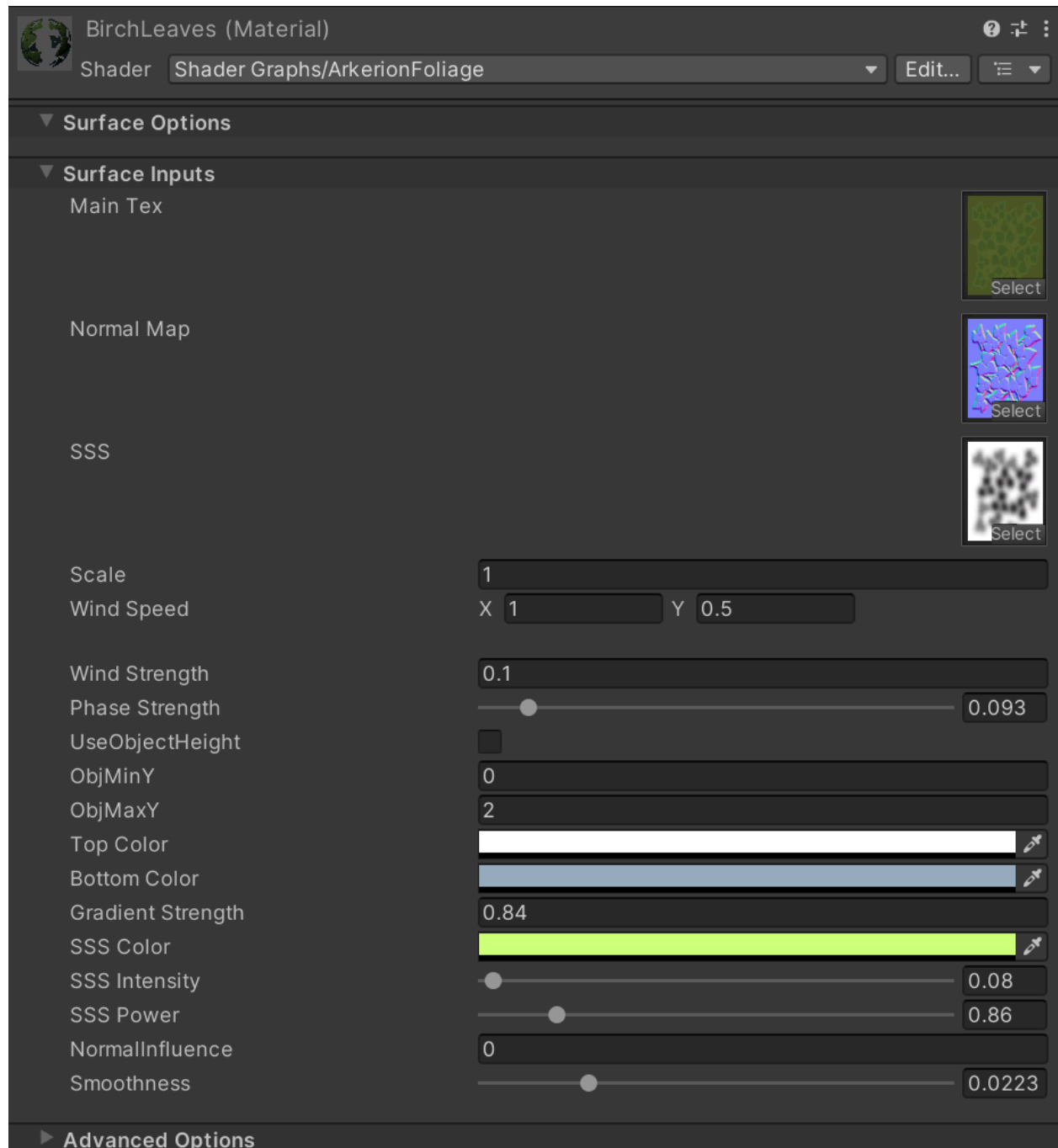


Arkerion Shaders - Usage Guide

ArkerionFoliage Shader (Mesh-based)



Unity version: 2022.3 LTS (URP)

Render Pipeline: URP
Shader type: Lit (Shader Graph)

Contents

Mesh-based foliage sway shader (vertex animation)
Wind animation based on world-space noise
Supports vertex color masking (bottom → top)

Optional textures support:

Albedo (Main Tex)
Normal Map
SSS / Thickness texture

Optional vertical color gradient
Custom Subsurface Scattering (back scattering) logic
Demo scene with example foliage meshes (grass & tree leaves)

Key Features

No scripts required
Fully Shader Graph based
Artist-friendly material parameters

Works with:

simple “flat color” foliage
textured foliage (albedo + normal + SSS)
Suitable for grass, leaves, bushes, small plants

How to use

Import the provided .unitypackage
Open the demo scene (if included)
Assign the shader to a foliage material
Apply the material to a mesh
Make sure the mesh has vertex colors (see below)
Adjust wind, color and SSS parameters in the material

Vertex Color Usage (Important)

The shader uses vertex color to control wind influence:
Black (0) — no movement (base of foliage)
White (1) — full movement (tips)
A vertical gradient from bottom → top is recommended

This allows:

stable roots

moving tips

natural-looking wind animation

Texture Support (Optional)

The shader can work with or without textures.

Main Tex (Albedo)

Acts as the base color texture

Can be used alone (no gradient)

Used as the base when gradient is disabled

Normal Map

Standard tangent-space normal map

Enhances lighting and surface depth

Normal Influence controls its strength

SSS Texture (Thickness)

Grayscale texture

White = thin areas (stronger scattering)

Black = thick areas (less scattering)

Typically used for leaves

 All textures are optional.

The shader also works with flat colors only.

Color Gradient (Optional)

The shader supports an optional vertical color gradient.

Gradient Strength

0 — gradient disabled (pure albedo / Main Tex)

1 — full gradient effect

When gradient is disabled, the shader correctly falls back to Main Tex / Albedo, without darkening artifacts.

Top Color

Applied to upper parts of foliage

Usually brighter / warmer

Bottom Color

Applied to lower parts
Usually darker to ground the foliage

Subsurface Scattering (Back Scatter)

Custom back-scattering logic is implemented to simulate light passing through thin foliage.

SSS Color

Tint of the scattering light
Best results with yellow-green tones

SSS Intensity

Controls overall scattering strength
0 disables SSS completely

SSS Power

Controls the sharpness of scattering
Lower values - softer glow
Higher values - tighter highlights
Clamped to 0–5 for stability

Wind Parameters

Scale

Controls wind noise scale.
Higher values - larger, slower waves
Lower values - smaller, faster motion
Recommended: 0.5 – 2

Wind Speed (X / Y)

Controls wind direction and speed in world space.
X — world X axis
Y — world Z axis
Using different values helps avoid uniform motion.

Wind Strength

Overall movement amplitude.
Recommended: 0.05 – 0.2

Phase Strength

Controls desynchronization between instances.
Lower - synchronized motion
Higher - organic, random motion
Recommended: 0.05 – 0.3

Lighting Parameters

Normal Influence

Controls how strongly normals affect lighting.

High values can look unnatural on foliage

Smoothness

Controls surface smoothness.

Foliage looks best with very low values

Recommended range: 0 – 0.1

Higher values may cause plastic-like highlights

Notes

Shader is designed for Lit pipeline, so lighting affects shading

Bottom darkening is physically correct and comes from lighting, not color logic

Can be used for LOD-friendly foliage meshes

Suitable as a base foliage shader for production

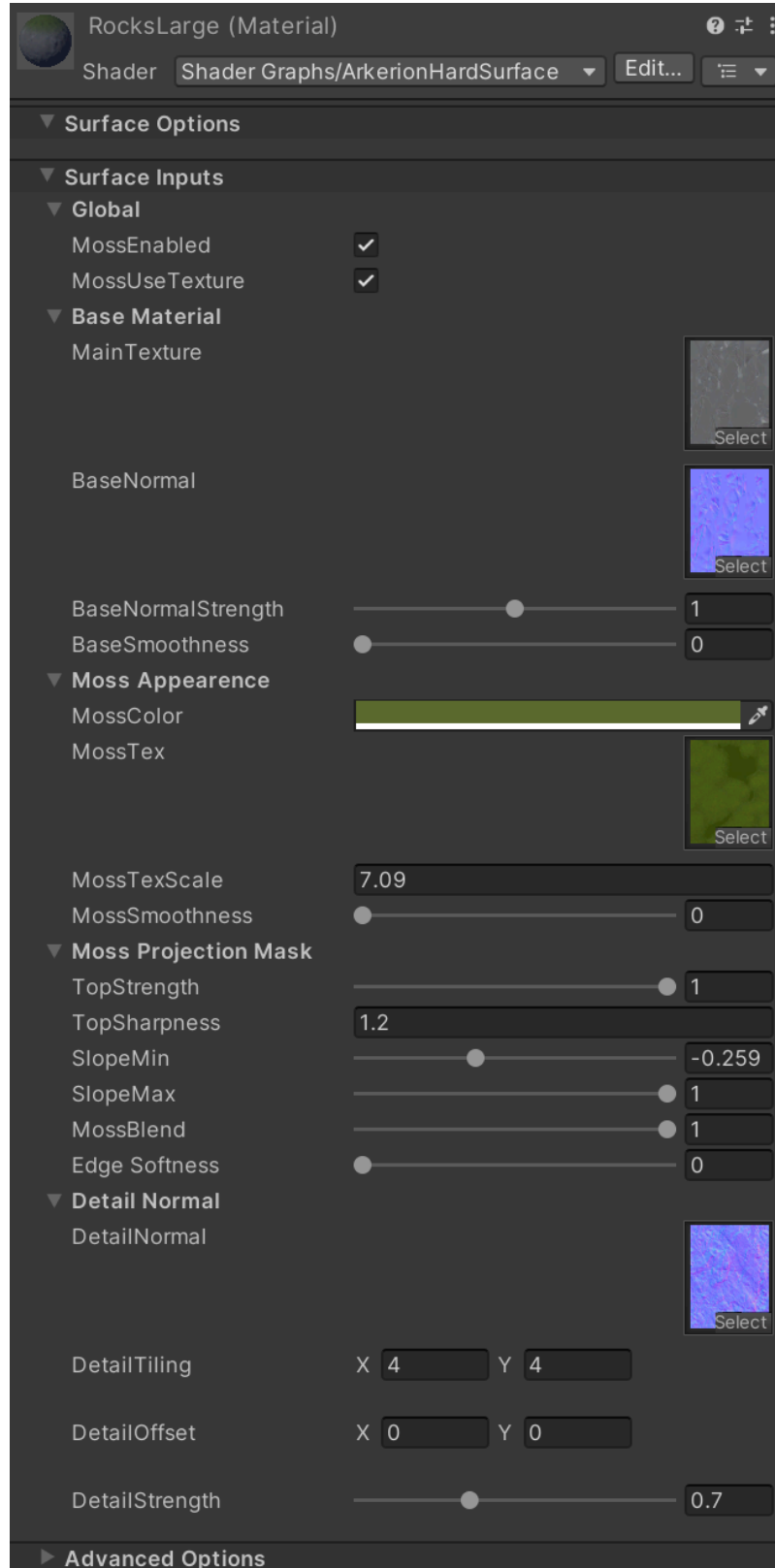
Summary

This shader supports both:

simple stylized foliage

more realistic textured foliage with normals and SSS

ArkerionHardSurface Shader (URP)



Unity version: 2022.x
Render Pipeline: URP
Shader type: Lit (Shader Graph)

Contents

Top-projected moss shader (world-space)
Slope-based moss masking
Color or texture-based moss
Edge-softened moss blending
Detail normal support
Global moss override system
Separate surface smoothness control

Description

Moss Projection Shader for Unity URP, inspired by the Unreal Engine reference material.
Designed to add moss to upward-facing and sloped surfaces with flexible artistic control.

The shader is intended for environment assets such as rocks, trees, cliffs, stumps, and terrain props.

It is fully material-instance driven and does not require per-object setup.

Key Features

Top-down moss projection (world-space)
Slope-based masking (top & angle control)
Switch between moss color and moss texture
Edge softening for natural moss borders
Detail normal map with tiling & offset
Separate smoothness control for base and moss
Optional global override (affects all materials)
No custom scripts required for basic usage

Shader Parameters

Moss Control

MossEnabled - Enables or disables moss
TopStrength - Overall influence of top projection
TopSharpness - Sharpness of the top mask
SlopeMin / SlopeMax - Controls slope range where moss appears
EdgeSoftness - Softens moss edges to avoid hard cutoffs

Moss Appearance

MossColor - Tint color for moss
MossTex - Optional moss texture
MossUseTexture - Switch between color or texture
MossTexScale - Moss texture tiling
MossBlend - Blending strength with base material

Normal & Detail

Base Normal - Uses original model normal together with moss and detail normals
BaseNormalStrength - Intensity of base normal before blending
DetailNormal - Detail normal map for moss
DetailTiling - UV tiling for detail normal
DetailOffset - UV offset for detail normal
DetailStrength - Intensity of detail normal

Surface Properties

BaseSmoothness - Smoothness of base material
MossSmoothness - Smoothness of moss layer

Global Override

The shader supports global parameters, allowing you to:

Enable or disable moss globally
Override moss color or moss texture across all materials
Perform fast art-direction changes or seasonal variations

How to Use

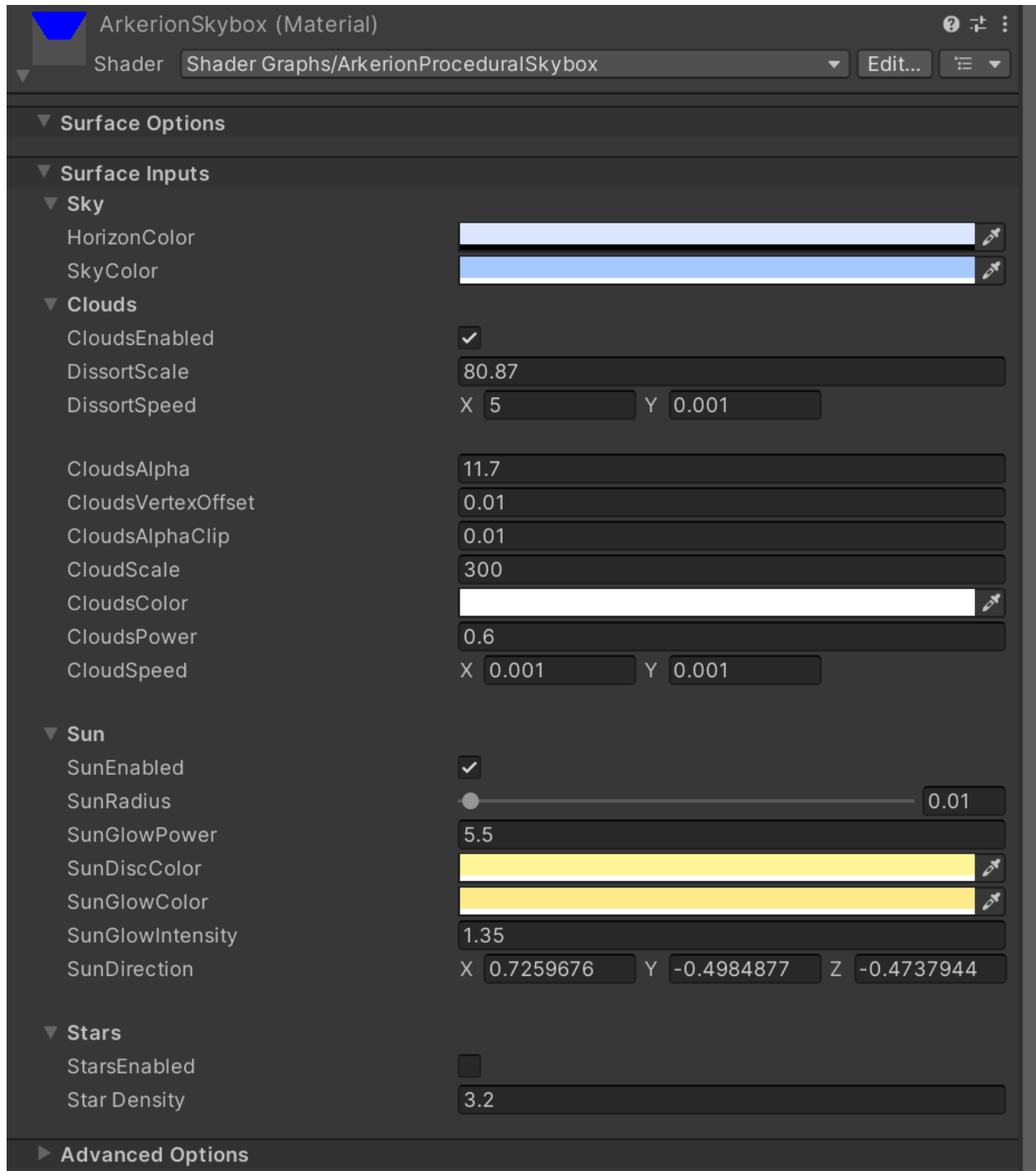
Assign the shader to a Lit material
Apply the material to environment meshes
Adjust top projection and slope values

Choose between moss color or moss texture
Tune blending, normals, and smoothness as needed

Usage Notes

Works best on assets with properly oriented normals
Moss texture is optional - color-only mode is fully supported
Designed for URP Lit workflow
Fully material-instance driven

ArkerionProceduralSkybox (URP, Unity 2022)



Unity version: 2022 LTS (URP)

Render Pipeline: URP

Shader type: Shader Graph (Custom Mesh Sky)

Contents

Custom stylized sky solution using a mesh instead of Unity Skybox

Procedural sky gradient

Procedural sun and glow

Procedural stars

Procedural animated clouds

Runtime sun synchronization via helper script

Key Features

Fully procedural (no baked skybox)

Mesh-based sky (no Unity Skybox dependency)

Artist-friendly material parameters

Performance-friendly design

Includes:

Sky Gradient

Sun

Stars

Procedural Clouds

What's Included

Sky Gradient

Procedural vertical sky gradient with customizable colors.

Sun

Procedural sun disc and glow.

Sun position is driven by the Directional Light.

Can be enabled or disabled.

Stars

Procedural star field.

Can be enabled or disabled independently.

Procedural Clouds

Fully procedural (no baked skybox).

Animated and layered for depth.

Designed to blend naturally with the sky.

Can be faded out or fully disabled via intensity control.

Runtime Sun Synchronization (Script)

The project includes a small helper script attached to a GameObject in the scene.

Purpose:

Automatically synchronizes the sky shader with the active Directional Light.
Updates sun direction in the shader in real time.

Notes:

Works in Play Mode and Edit Mode.

Requires:

- * Reference to the sky material.
- * A Directional Light (or `RenderSettings.sun`).
- No expensive per-frame operations.

Rotating the Directional Light automatically updates the sun position in the sky.

How to use

Assign the sky material to the provided mesh.
Assign a Directional Light.
Ensure the sun synchronization script is active.
Adjust visual settings via the material inspector.

Most parameters are visually driven and safe to tweak.

Technical Notes

Uses a custom curved mesh instead of Unity Skybox.
Built for URP.
Designed to work without HDR by default for better performance.
HDR can be enabled if higher color depth is required.

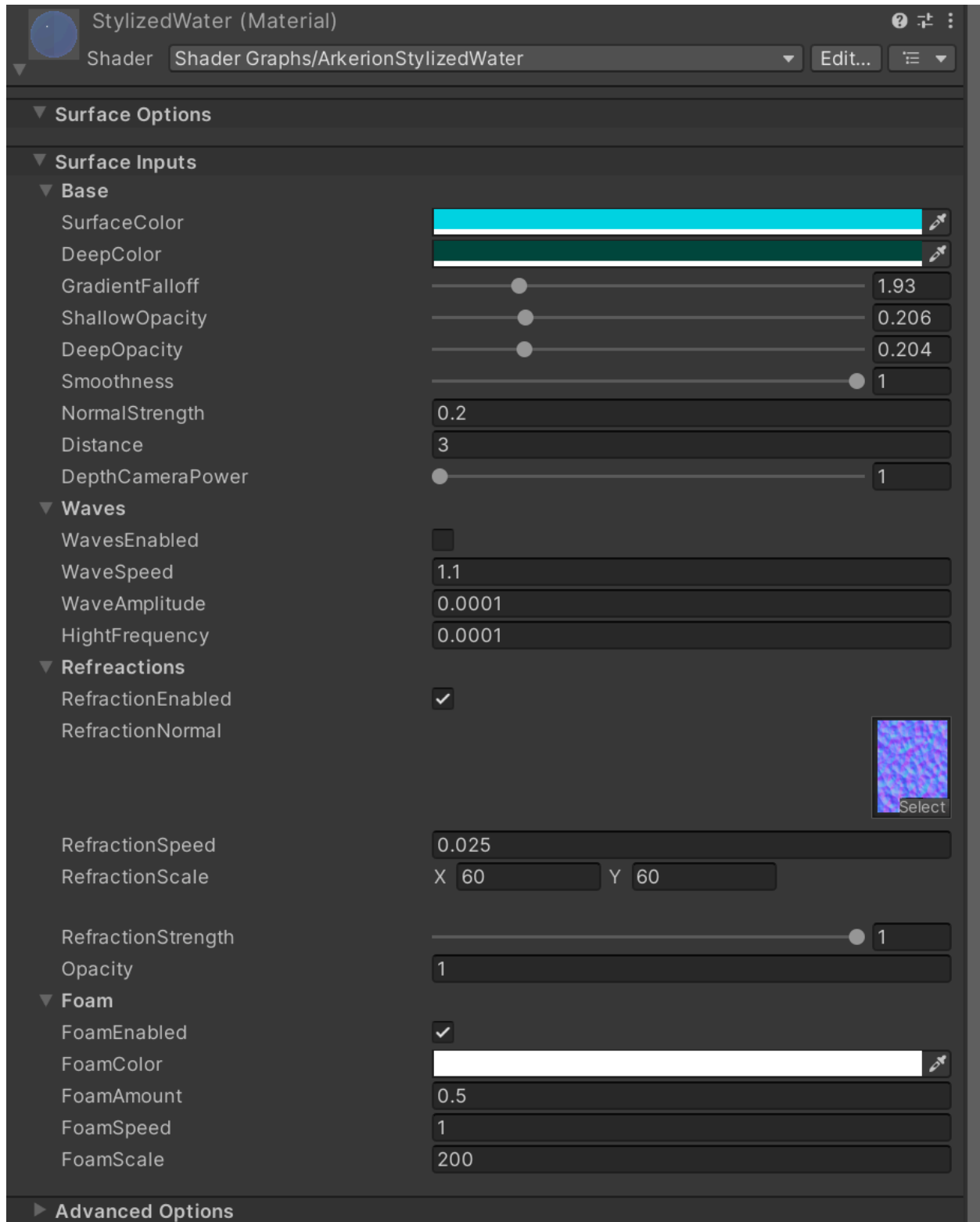
Performance

No volumetric rendering.
No raymarching.
No post-processing dependency.
Suitable for PC and mobile-friendly setups.

Notes

This shader was built iteratively and visually tuned.
Not every internal parameter is documented by design.
The system is meant to be artist-friendly and exploratory.

STYLIZED WATER SHADER (URP)



Unity version: 2022.x

Render Pipeline: URP
Shader type: Lit (Stylized usage)

Contents

Stylized procedural water shader
Depth-based color blending
Vertex wave animation
Foam at object intersections
Fake reflections
Animated normals
Performance feature toggles

Description

Stylized Water Shader for Unity URP (Unity 2022).
Optimized for PC and Mobile.
Designed for ponds and lakes (not ocean).

The shader is fully procedural and built with future expansion in mind
(river, waterfall, underwater effects).

Key Features

Depth-based color gradient (shallow / deep colors)
Vertex waves (toggle on/off)
Foam at object intersections (depth-based)
Fake reflections (Fresnel-based, lightweight)
Animated normal map (2 layers, same texture)

Performance toggles:
Waves Enabled
Foam Enabled
Reflections Enabled

Shader Parameters

Water:
SurfaceColor
DeepColor
Distance (depth control)

Waves:
WaveSpeed
WaveAmplitude

HighFrequency
WavesEnabled

Foam:

FoamColor
FoamAmount
FoamSpeed
FoamScale
FoamEnabled

Reflections:

ReflectionColor
ReflectionStrength
ReflectionPower
ReflectionsEnabled

Normals:

NormalStrength
Normal Map (single texture, 2 animated layers)

How to use

Assign the material to a flat water mesh.
Adjust depth colors and distance for desired look.
Enable or disable Waves, Foam, and Reflections as needed.
Tune parameters depending on target platform (PC/Mobile).

Technical Details

HDR: Not required (LDR friendly)
Mobile-friendly (no real-time reflections, no SSR)

Performance Notes

No planar reflections or screen-space reflections
Fake reflections are Fresnel-based and very cheap
All heavy features can be disabled via toggles
Suitable for low-end mobile devices